

IoT Garden Monitor · Premium Course

2-session series (2 hrs each) · 10–14 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Build a sensor kit that keeps a real seedling alive, then automate its care.

DETAIL

Track	Arduino Robotics & Electronics
Format	2-session series (2 hrs each)
Ages	10–14 yrs
Level	Intermediate
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Parts pack	Optional, KES 1,000: Soil-moisture sensor, Mini water pump + tubing, Pot and seedling
Standalone	No
Prerequisites	Temperature logger (free, recommended)

Programme

1. **Sense.** Soil moisture sensor + DHT11 (humidity/temp) + LDR (sunlight), with readings on an LCD. Plant a real seedling.
2. **Automate.** Set thresholds — if the soil is dry a relay triggers a small water pump; a buzzer alerts when conditions are off.

What they take home

 Sensor kit + a seedling in a monitored pot

Why parents love it

“STEM meets nature. They build tech that actually keeps a plant alive.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**IoT Garden Monitor**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.